

Example: Laplace transform, frequency-shift theorem**1 Problem**

Given

$$f(t) = \cos(3t) - e^{-4t} \sin(5t),$$

find

$$F(s).$$

2 Solution

It is implied that $F(s) = \mathcal{L}[f(t)]$.

$$F(s) = \mathcal{L}[f(t)] \tag{1}$$

$$\mathcal{L}[\cos(3t) - e^{-4t} \sin(5t)] \tag{2}$$

$$\mathcal{L}[\cos(3t)] - \mathcal{L}[e^{-4t} \sin(5t)] \tag{3}$$

$$\frac{s}{s^2 + 3^2} - \frac{5}{(s + 4)^2 + 5^2} \tag{4}$$